The Contact Problem

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Contents

1 The Contact Problem

1.1 Description and Connection with Cold Spray

Contact problems describe the mechanical interaction between two or more deformable bodies that come into contact. The primary challenge is determining the contact interface while enforcing the non-penetration constraint. Depending on the physical model, frictional effects may also be incorporated to describe tangential resistance and sliding along the contact surface.

In cold spray additive deposition, metallic particles impact a target substrate at supersonic velocities, producing severe plastic deformation and localized adiabatic heating. These conditions promote adiabatic shear instability amongst other effects, which allows the particle to then metallurgically bond to the substrate material. [1] Particle-to-subtrate and particle-to-particle bonding are both present in cold spray deposition, which may be of different materials. Determination for the correct physics locally necessarily requires the contact problem.

1.2 The Mathematical Problem

Entire textbooks are devoted to contact mechanics; therefore, only a brief overview is presented here. [3]

2 Elastodynamics

The cold spray process is necessarily a dynamic one, so one must resolve the physics incrementally.

2.1 Linear Elasticity

One may model the cold spray process as an elastic-plastic problem, though the elasticity portion is far less significant.

The strong form of linear elasticity is given by Cauchy's equations.

$$\rho\ddot{x} = \nabla \cdot \sigma + f \tag{1}$$

$$\sigma = \lambda \operatorname{tr}(\varepsilon)\mathbf{I} + 2\mu\epsilon \tag{2}$$

$$\varepsilon = \frac{1}{2}(\nabla x + (\nabla x)^T) \tag{3}$$

We will now convert this into the weak form that will be more useful for us which will be elaborated upon in the next section. v will be our test functions.

$$\int \rho\ddot{x} \cdot v = \int (\nabla \cdot \sigma) \cdot v + \int f \cdot v \tag{4}$$

$$\int \rho\ddot{x} \cdot v + \int \sigma(x) : \varepsilon(v) = \int f \cdot v \tag{5}$$

By application of the divergence theorem on the internal stress term.

We expand the stress and strain tensor with the constitutive equations 2 and 3.

$$\int \rho\ddot{x} \cdot v + \int \lambda(\boldsymbol{\nabla} \cdot x)(\boldsymbol{\nabla} \cdot v) + 2\mu\epsilon(x) : \epsilon(v) = \int f \cdot v \tag{6}$$

Naturally, we convert the equation into Galerkin's formulation to then input into FreeFEM. Einstein summation convention here will be used.

$$x(t) = X_i(t)u_i(x) \tag{7}$$

$$\rho\ddot{X}_i \int u_i \cdot v_j + X_i \int \lambda(\boldsymbol{\nabla} \cdot u_i)(\boldsymbol{\nabla} \cdot v_j) + 2\mu\varepsilon(u_i) : \varepsilon(v_j) = \int f \cdot v_j \tag{8}$$

Observe that this gives us the following equation.

$$M_{ij}\ddot{x}_j + K_{ij}x_j = f_i \tag{9}$$

$$M_{ij} = \int u_j \cdot v_i \ ; \ K_{ij} = \int \lambda(\boldsymbol{\nabla} \cdot u_j)(\boldsymbol{\nabla} \cdot v_i) + 2\mu\varepsilon(u_j) : \varepsilon(v_i) \ ; \ f_i = \int f \cdot v_i \tag{10}$$

Which is the form to then plug into the α integrators such as HHT-α and Newark-β.

2.2 Newark Beta

Here we will present the incremental Newark Beta integrator for structural dynamics.[2]

We follow Gavin, H.'s lecture notes on incremental formulation of the Newark integrator.

The Newark integrator begins with the following assumptions on the form of the differential equation and the discretizations.

$$M\ddot{x} + C\dot{x} + Kx + R(x, \dot{x}) = f \tag{11}$$

$$\dot{x}_{i+1} \approx \dot{x}_i + h\left[(1-\gamma)\ddot{x}_i + \gamma\ddot{x}_{i+1}\right] \tag{12}$$

$$x_{i+1} \approx x_i + h\dot{x}_i + h^2\left[\left(\frac{1}{2} - \beta\right)\ddot{x}_i + \beta\ddot{x}_{i+1}\right] \tag{13}$$

Where M is the mass matrix, C the dissipation matrix, K the stiffness matrix, R will be any nonlinear components, f being any external forces, and A_i being $A(t = ih)$.

The free parameters β and γ are left to be tuned for desired integrator behaviors. Choices such as $\gamma \leq 1/2$, the integrator is conditionally stable, whilst the chosen method for this project is implicit and unconditionally stable with $\beta = 1/4$ and $\gamma = 1/2$.

We now solve for $\ddot{x}_{i+1}$.

$$\ddot{x}_{i+1} = \frac{1}{\beta h^2}(x_{i+1} - x_i) - \frac{1}{\beta h}\dot{x}_i - \left(\frac{1}{2\beta} - 1\right)\ddot{x}_i \tag{14}$$

Giving us this formula for the next velocity.

$$\dot{x}_{i+1} = \frac{\gamma}{\beta h}(x_{i+1} - x_i) - \left(\frac{\gamma}{\beta} - 1\right)\dot{x}_i + h\left(1 - \frac{\gamma}{2\beta}\right)\ddot{x}_i \tag{15}$$

2.3 Numerical FreeFEM Implementation

We consider a linear elastic block resting on a table, slumping under the effect of gravity. The constant parameters are as follows.

Name	Notation in Code	Value
Gravity	g	-9.8 m/s^2
Density	ρ	1
Young's modulus	E	1E3 $kg/m/s^2$
Poisson modulus	nu	0.3
Newark parameter	beta	0.25
Newark parameter	gamma	0.5
Rayleigh dissipation parameter	AlphaM	0.01
Rayleigh dissipation parameter	AlphaK	0.01

We introduce some notation to simplify the equations 14 and 15.

$$c_0 = \frac{1}{\beta h^2} \; ; \; c_1 = \frac{1}{\beta h} \; ; \; c_2 = \frac{1}{2\beta} - 1 \tag{16}$$

$$c_3 = \frac{\gamma}{\beta h} \; ; \; c_4 = \frac{\gamma}{\beta} - 1 \; ; \; c_5 = h\left(1 - \frac{\gamma}{2\beta}\right) \tag{17}$$

As shown in the code below.

```
// Beta Newmark constants
real c0 = 1./(beta*dt*dt);
real c1 = 1./(beta*dt);
real c2 = 1./(2.*beta) - 1.;
real c3 = gamma/(beta*dt);
real c4 = gamma/beta - 1.;
real c5 = dt*(gamma/(2*beta) - 1.);
```

The Lame parameters for the stress tensor are given as follows.

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \; ; \; \mu = (3 * 10^{-2})\frac{E}{2(1+\nu)} \tag{18}$$

Where we have chosen to scale the 2nd Lame parameter by 3e-2 to amplify visual effects.

A particular sanity check is that udner dampening, the system will converge to the steady state, which one can trivially compute. So, we add in the Rayleigh dissipation terms into the dissipation matrix C.

$$C = \alpha_M M + \alpha_K K \tag{19}$$

Here we introduce some more notations to simplify the Galerkin formulation.

$$M(a) = \int a \cdot v \; ; \; K(a) = \int \lambda(\boldsymbol{\nabla} \cdot a)(\boldsymbol{\nabla} \cdot v) + 2\mu\varepsilon(a) : \varepsilon(v) \tag{20}$$

As shown in the code below.

```
// Macros M and K as described in the PDF
macro M(a) (a[0]*pi+a[1]*pj) //
macro K(a) (lambda*(div(a)*div([pvi,pvj]))+2*muK*mu*eps(a)'*
    eps([pvi,pvj])) //
```

Altogether, we form the Galerkin formulation.

$$\begin{aligned}
M\ddot{x}_{i+1} + C\dot{x}_{i+1} + Kx_{i+1} - f_{i+1} = 0 \\
M(c_0(x_{i+1} - x_i) - c_1\dot{x}_i - c_2\ddot{x}_i) + \\
(\alpha_M M + \alpha_K K)(c_3(x_{i+1} - x_i) - c_4\dot{x}_i - c_5\ddot{x}_i) + \\
Kx_{i+1} - f_{i+1} = 0 \\
c_0 M(x_{i+1}) - c_0 M(x_i) - c_1 M(\dot{x}_i) - c_2 M(\ddot{x}_i) + \\
\alpha_M(c_3 M(x_{i+1}) - c_3 M(x_i) - c_4 M(\dot{x}_i) - c_5 M(\ddot{x}_i)) + \\
\alpha_K(c_3 K(x_{i+1}) - c_3 K(x_i) - c_4 K(\dot{x}_i) - c_5 K(\ddot{x}_i)) + \\
K(x_{i+1}) - f_{i+1=0}
\end{aligned}$$

Finally, we shuffle terms to have all additions in 1 integral and all subtractions in another.

$$\begin{aligned} c_0 M(x_{i+1}) + \alpha_M c_3 M(x_{i+1}) + \alpha_K c_3 K(x_{i+1}) + K(x_{i+1}) \\ - \\ c_0 M(x_i) + c_1 M(\dot{x}_i) + c_2 M(\ddot{x}_i) + \\ \alpha_M (c_3 M(x_i) + c_4 M(\dot{x}_i) + c_5 M(\ddot{x}_i)) + \\ \alpha_K (c_3 K(x_i) + c_4 K(\dot{x}_i) + c_5 K(\ddot{x}_i)) + \\ f_{i+1} = 0 \end{aligned}$$

As shown in the code.

```
problem BetaNewark([px,py],[pvi,pvj])
= int2d(Ph)(
        c0*M([px,py])+
                AlphaM*c3*M([px,py])+
                AlphaK*c3*K([px,py])+
                K([px,py])
        )
- int2d(Ph)(
                c0*M([un,vn])+
                c1*M([uvn,vvn])+
                c2*M([uan,van])+
                AlphaM*(c3*M([un,vn])+c4*M([uvn,vvn])+c5*M([
                    uan,van]))+
                AlphaK*(c3*K([un,vn])+c4*K([uvn,vvn])+c5*K([
                    uan,van]))+
                g*pvj
        )
+ on(bottom, py=0);
```

Using FreeFEM's builtin hTriangle function, we can obtain the size of the finite elements. This way we can adequately choose the time step size for a good CFL number. The minimum size element in this case has the size of 0.0820061m.

Wave Type	Speed	Time Step for CFL=1
Pressure	24.4949 m/s	0.00334788s
Shear	3.39683 m/s	0.0241419s

So we choose a conservative time step of 0.002s.

2.4 Results

The results can be viewed as a time series in Paraview, or any .vtu viewer. The block jiggles and the movements slowly decay away.

3 FreeFEM Implementation

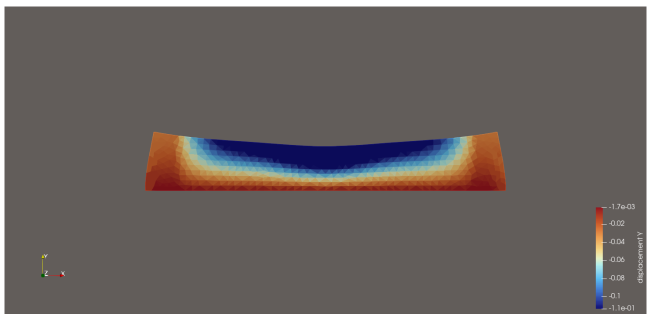
References

[1] H. Assadi, H. Kreye, F. Gärtner, and T. Klassen. Cold spraying – a materials perspective. *Acta Materialia*, 116:382–407, 2016.

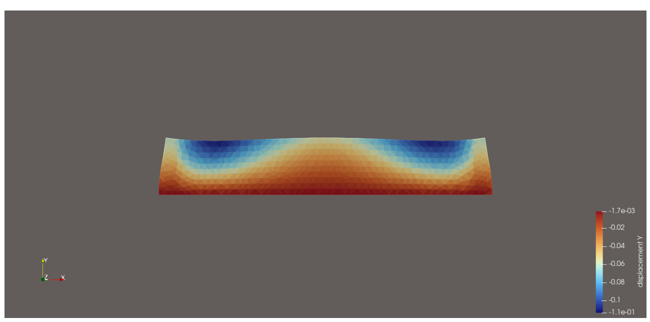
[2] Henri P. Gavin. Numerical integration in structural dynamics. Course notes for CEE 541: Structural Dynamics, 2020. Fall 2020.

[3] Peter Wriggers. *Computational Contact Mechanics*. Springer, Berlin, Heidelberg, 2nd edition, 2006.

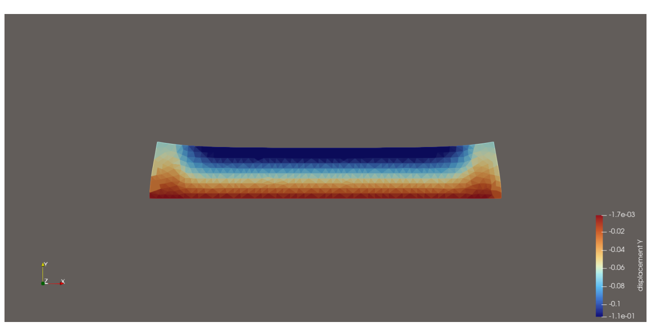
(a) t=0s



(b) t=0.75s



(c) t=1.25s



(d) t=7.5s